

Design Note — Assignment 2: Learning from Click-Logs on MIND and EB-NeRD

CS4.406 Information Retrieval & Extraction · Parth Dhawale & Peeyush Prashant

Code: <https://github.com/Zenith053/Assignment1-IRE> (branch `parth/q4-serving`)

1 What we built

A two-stage news recommender on MIND-small and EB-NeRD *small* (EB-NeRD was rebuilt from demo to small for A2, $\sim 10\times$ more data). **Stage 1** (A1’s candidate generator) retrieves up to 200 candidates per request: BM25 top-80 $\cup$ FAISS top-100 over A1 embeddings (MiniLM for MIND, the provided word2vec for EB-NeRD) $\cup$ train-popularity top-20. **Stage 2** re-ranks them. We built two families of re-rankers: a GBDT/MLP over hand-crafted behavioural features (Q1–Q2) and NRMS [1], the official baseline, plus a popularity/freshness extension (Q3). Every evaluation is on a strict temporal split (MIND train 2019-11-09 – 2019-11-12, test 2019-11-15 – 2019-11-15; EB-NeRD train 2023-05-18 – 2023-05-23, test 2023-05-25 – 2023-06-01); each split’s user histories come from a snapshot that ends before the split starts, asserted by tests (`tests/test_no_leakage.py`). Two evaluation *universes* are kept apart: **A** re-ranks the impression’s own shown (inview) list, which is what Codabench scores, and **B** is the full pipeline: retrieve 200 from the catalogue, then re-rank.

Table 1: Data used throughout (temporal splits; the shipped held-out file is the test split).

dataset	train imp.	val imp.	test imp.	articles	cand./imp.	test users <20 clicks
MIND-small	95,071	61,894	73,152	65,238	23	59%
EB-NeRD small	192,884	40,003	244,647	20,738	9	16%

2 Q1: Click-history, session and article features

Table 2: Q1 feature table. ^EEB-NeRD only (MIND has no timestamps, sessions or publish times); on MIND these are NaN, never faked.

group	features (30 in total)
Click history	history length (raw, log), categories, category entropy, decay-weighted mass; hours since last click ^E , mean past dwell time and scroll ^E
Session	candidate position (raw, normalised), inview size, hour of day, day of week; index in session, clicks so far, seconds since session start ^E
Article	train clicks (log), popularity rank, is-head, trailing clicks 24 h / 168 h , title length; freshness (hours since publish) ^E
Match	category match, category match rate; BM25 and semantic score and their in-impression ranks, popularity rank in impression

Behaviour-window boundary. Two new mechanisms read *other* impressions, so each has its own test: trailing click counts count only clicks with timestamp strictly before the impression (a counterfactual test rebuilds the counter from past-only events and demands identical counts), and session features are asserted strictly increasing in real time. Columns only known after the click (current `read_time`, `scroll_percentage`, lifetime `total_pageviews` ...) are excluded from every feature; a user’s *past* dwell time is an ordinary feature. The pre-computed **hybrid** score is excluded because it was fitted on the validation split that also early-stops the re-rankers.

3 Q2: Retrieve-then-rank re-rankers

Architecture. *Option A:* LightGBM with a LambdaRank objective [6, 7] (optimises nDCG directly; 31 leaves, ≥ 30 samples per leaf, L2 1.0, feature fraction 0.7, early stopping on validation nDCG@10). *Option B:* a small

MLP (64-32) trained with a listwise softmax per impression, with a missing-value indicator per feature so NaN features are usable. Both train on the train split, early-stop on validation, and report on test. Raw train-click counts were replaced by in-impression ranks after they dominated the GBDT’s splits (a high-cardinality bias) and made it lose to semantic.

Table 3: Q2, Universe A (re-rank the shown list), test metrics on 10,000 impressions per dataset.

	MIND AUC	MRR	nDCG@10	EB AUC	MRR	nDCG@10
A1 popularity	0.4955	0.2182	0.2836	0.4441	0.2629	0.3839
A1 BM25	0.5702	0.2658	0.3445	0.5204	0.3292	0.4450
A1 semantic (before)	0.6443	0.3140	0.4033	0.5296	0.3370	0.4538
GBDT (after)	0.6220	0.3013	0.3865	0.7499	0.5226	0.6208
MLP (after)	0.5907	0.2796	0.3524	0.6065	0.3684	0.4946

Table 4: Q2, after vs before (best single A1 scorer, semantic on both): paired bootstrap 95% CI, 10,000 resamples; * = CI excludes zero. MRR and nDCG@5 move the same way (all significant).

	MIND Δ AUC	MIND Δ nDCG@10	EB Δ AUC	EB Δ nDCG@10
GBDT – semantic	−0.022 [−0.026, −0.018]*	−0.017 [−0.020, −0.014]*	+0.220 [+0.213, +0.228]*	+0.167 [+0.160, +0.174]*
MLP – semantic	−0.054 [−0.061, −0.046]*	−0.051 [−0.057, −0.045]*	+0.077 [+0.069, +0.085]*	+0.041 [+0.034, +0.047]*

Table 5: Q2, Universe B (retrieve top-200, then re-rank), 3,000 test impressions each. Accuracy is conditional on a click being retrieved; end-to-end = recall $\times$ conditional.

	MIND	EB-NeRD small
recall@200 (click retrieved)	0.235	0.096
AUC retrieved, stage-1 order	0.673 [0.657, 0.689]	0.444 [0.413, 0.479]
AUC retrieved, GBDT	0.494 [0.477, 0.512]	0.917 [0.896, 0.935]
nDCG@10 retrieved, before $\rightarrow$ after	0.060 $\rightarrow$ 0.012	0.017 $\rightarrow$ 0.409
end-to-end nDCG@10, before $\rightarrow$ after	0.0141 $\rightarrow$ 0.0029	0.0016 $\rightarrow$ 0.0393

Observations. Re-ranking over behavioural features *loses* to A1’s semantic scorer on MIND (significant on every metric) but *wins* by +0.22 AUC on EB-NeRD. Top GBDT features (gain): MIND `train_clicks_log1p` 0.19, `semantic` 0.15, `popularity_rank` 0.10; EB-NeRD `rolling_clicks_24h` 0.27, `freshness_hours` 0.13, `rolling_clicks_168h` 0.09. EB-NeRD is driven by *what is being clicked now* and *how new the article is* — the signals Q3 adds to NRMS. Stage 1 caps the whole system: 76% (MIND) and 90% (EB-NeRD) of test clicks are never retrieved. On retrieved lists the GBDT trained on shown negatives is at chance on MIND (the negatives’ distribution shifts) but near-perfect on EB-NeRD, where freshness and trailing clicks separate the one live article from mostly stale retrieved ones. Making Q2 run required pinning LightGBM and FAISS to one thread (faiss, torch and LightGBM each link an incompatible OpenMP runtime; multi-threaded training segfaulted), and fixing a metric bug that had applied recall twice in Universe B.

4 Q3: Baseline reproduced, then beaten

Baseline: NRMS [1], the official ebnerd-benchmark baseline, ported to PyTorch with the benchmark’s hyperparameters: title length 30, history 20, 4 sampled negatives per click, 20 heads $\times$ 20 dims, additive attention 200, dropout 0.2, Adam 10^{-4} , batch 32, trainable word embeddings initialised from XLM-RoBERTa-base [5] (one multilingual vocabulary for Danish and English). A news encoder (multi-head self-attention over title tokens, additive pooling) and a user encoder (the same over the last 20 clicked articles) give 400-d vectors; the score is their dot product. Deviations: PyTorch port; padding masked out of every softmax (59% of MIND test users have <20 clicks); evaluated on our temporal split, so numbers are not the published ones. NRMS is *below* A1 semantic on MIND (Δ AUC −0.0186 [−0.0209, −0.0164]*) and *above* it on EB-NeRD (+0.0236 [+0.0218, +0.0253]*), where A1 only had the weaker provided word2vec vectors.

Principled change: popularity- and freshness-aware NRMS (after PP-Rec [2]). NRMS reads only titles, but news decays within hours and Q2 showed trailing clicks and article age are EB-NeRD’s strongest signals. Each candidate gets $\log(1 + \cdot)$ of its clicks in the trailing 1 h / 24 h / 7 d (strictly before the impression;

the vectorised counter is tested equal to Q1’s causal `RollingPopularity`) and, on EB-NeRD, its age. A small MLP turns them into a signal score, blended with the content score by a per-user gate $g = \sigma(w^\top u + b)$: $s = (1-g) s_{\text{content}} + g s_{\text{signal}}$; the ablation replaces the gate with a plain sum. The variant reported as “improved” is chosen by mean *validation* AUC among pre-stated candidates (MIND: plain sum 0.6348 vs gate 0.6317), never by test.

Table 6: Q3 baseline vs improved on the full test splits, 3 seeds each (MRR and nDCG are seed means).

	model	test AUC, seeds 13 / 14 / 15	AUC mean $\pm$ sd	MRR	nDCG@5	nDCG@10
MIND-small	NRMS (baseline)	0.6232 / 0.6217 / 0.6261	0.6237 $\pm$ 0.0022	0.2959	0.3214	0.3835
	improved: <code>nrms_pop_sum</code>	0.6359 / 0.6190 / 0.6314	0.6288 $\pm$ 0.0088	0.2972	0.3236	0.3860
EB-NeRD small	NRMS (baseline)	0.5539 / 0.5488 / 0.5498	0.5508 $\pm$ 0.0027	0.3471	0.3865	0.4645
	improved: <code>nrms_popfresh</code>	0.7246 / 0.7257 / 0.7265	0.7256 $\pm$ 0.0010	0.4860	0.5486	0.5928

Table 7: Q3 significance: paired bootstrap 95% CI (10,000 resamples) on each test impression’s metric averaged over the 3 seeds; * = CI excludes zero.

improved – NRMS		Δ AUC	Δ MRR	Δ nDCG@5	Δ nDCG@10
MIND-small	+0.0051 [+0.0043, +0.0060]*	+0.0013 [+0.0004, +0.0021]*	+0.0022 [+0.0013, +0.0032]*	+0.0025 [+0.0016, +0.0034]*	
EB-NeRD small	+0.1748 [+0.1733, +0.1761]*	+0.1389 [+0.1375, +0.1404]*	+0.1622 [+0.1607, +0.1636]*	+0.1283 [+0.1270, +0.1295]*	

Table 8: Q3 ablation, paired bootstrap 95% CI on the full test split; * = CI excludes zero.

ablation (EB-NeRD: seed 13)	Δ AUC	Δ MRR	Δ nDCG@10
EB-NeRD: + freshness only vs NRMS	+0.093 [+0.092, +0.094]*	+0.072 [+0.070, +0.073]*	+0.068 [+0.066, +0.069]*
EB-NeRD: + popularity only vs NRMS	+0.171 [+0.169, +0.172]*	+0.135 [+0.134, +0.137]*	+0.125 [+0.123, +0.126]*
EB-NeRD: + both, learned gate (full) vs NRMS	+0.171 [+0.169, +0.172]*	+0.138 [+0.137, +0.140]*	+0.127 [+0.126, +0.128]*
EB-NeRD: + both, plain sum vs NRMS	+0.170 [+0.168, +0.171]*	+0.138 [+0.137, +0.140]*	+0.127 [+0.126, +0.129]*
EB-NeRD: popularity only vs full	+0.000 [−0.000, +0.001]	−0.003 [−0.004, −0.002]*	−0.003 [−0.003, −0.002]*
EB-NeRD: freshness only vs full	−0.078 [−0.079, −0.077]*	−0.066 [−0.068, −0.065]*	−0.060 [−0.061, −0.059]*
EB-NeRD: plain sum vs learned gate	−0.001 [−0.002, −0.001]*	+0.000 [−0.000, +0.001]	+0.000 [−0.000, +0.001]
MIND: + popularity, learned gate vs NRMS (seed 13)	+0.005 [+0.004, +0.006]*	+0.000 [−0.001, +0.001]	+0.001 [+0.000, +0.002]*
MIND: plain sum vs learned gate (3 seeds)	+0.003 [+0.002, +0.003]*	+0.002 [+0.001, +0.002]*	+0.001 [+0.001, +0.002]*

Findings. (1) On EB-NeRD the change is large and robust: +0.175 AUC, significant on every seed and metric, seed spread ± 0.001 . Popularity carries almost all of it; freshness alone adds +0.09 but is largely redundant with popularity (no AUC gain on top of it, +0.003 MRR/nDCG). (2) Freshness is not monotone: click rate peaks at 2-4h (0.120) and falls to 0.013 for articles older than 30 days; “newer ranks higher” scores 0.501 AUC, the bump rule 0.663. (3) The per-user gate does not earn its keep: equal to a plain sum on EB-NeRD and slightly worse on MIND. (4) On MIND the gain is small (+0.005) and *not robust to training randomness*: seed 14 is significantly worse than its baseline, the seed spread (± 0.009) exceeds the gain, and validation disagreed with test on that seed. The paired CI covers test-impression sampling, not training noise.

5 Q5: Extended evaluation

Table 9: Q5, full two-stage pipeline (Universe B, 3,500 test impressions per dataset), all metrics. Accuracy is conditional on a retrieved click; diversity, novelty (bits) and coverage are over the top-10 shown.

	system	recall	AUC ret.	MRR ret.	nDCG@5 ret.	nDCG@10 ret.	diversity	novelty	coverage
MIND	stage-1 order	0.274	0.670 [0.655, 0.683]	0.070	0.055	0.072 [0.060, 0.085]	0.682	15.70	0.029
	two-stage GBDT	0.274	0.570 [0.556, 0.584]	0.032	0.015	0.022 [0.015, 0.029]	0.852	10.04	0.017
EB-NeRD	stage-1 order	0.127	0.493 [0.467, 0.520]	0.035	0.018	0.032 [0.020, 0.044]	0.116	11.99	0.073
	two-stage GBDT	0.127	0.845 [0.826, 0.863]	0.217	0.215	0.258 [0.230, 0.289]	0.136	12.28	0.076

Table 10: Q5 slices for the two-stage GBDT pipeline, with bootstrap 95% CIs. EB-NeRD has no cold users under the <5-click rule (every user has ≥ 5), so the lowest-history quartile stands in.

	slice	n	recall@200	AUC retrieved	nDCG@10 retrieved	diversity
MIND	cold (<5 clicks)	377	0.199	0.521 [0.461, 0.574]	0.013 [0.000, 0.029]	0.880
	warm	3,123	0.283	0.575 [0.558, 0.590]	0.023 [0.016, 0.030]	0.849
	lowest-history quartile	637	0.204	0.510 [0.468, 0.548]	0.014 [0.002, 0.027]	0.879
	other users	2,863	0.290	0.580 [0.564, 0.595]	0.023 [0.016, 0.031]	0.846
	head click	100	0.500	0.728 [0.660, 0.793]	0.187 [0.103, 0.279]	0.849
	tail click	3,400	0.268	0.562 [0.548, 0.578]	0.013 [0.008, 0.018]	0.853
EB-NeRD	cold (<5 clicks)	0	—	—	—	—
	warm	3,500	0.127	0.845 [0.826, 0.863]	0.258 [0.230, 0.289]	0.136
	lowest-history quartile	295	0.112	0.786 [0.701, 0.861]	0.235 [0.112, 0.370]	0.136
	other users	3,205	0.129	0.850 [0.830, 0.866]	0.260 [0.229, 0.293]	0.136
	head click	17	0.882	0.920 [0.857, 0.971]	0.356 [0.194, 0.530]	0.125
	tail click	3,483	0.124	0.843 [0.825, 0.861]	0.255 [0.226, 0.290]	0.136

Observations. Beyond-accuracy metrics move against accuracy. On MIND the two-stage GBDT recommends a *more* diverse but far *less* novel list than stage 1 alone (it learned to push popular articles), and covers fewer articles; on EB-NeRD it gains accuracy with little change in diversity. Head-click impressions retrieve far better than tail ones (MIND 0.500 vs 0.268, EB-NeRD 0.882 vs 0.124): much of stage 1’s recall comes from the popularity arm. Lowest-history users retrieve and rank worse on both datasets. Coverage is reported as a point estimate; head-click slices have 100 (MIND) and 17 (EB-NeRD) impressions, so their CIs are wide.

Codabench. The hidden test sets carry no clicks, so trailing click counts cannot be computed on them. We measured what that does on our labelled EB-NeRD test split: the best Q3 model scores 0.7246 with real trailing clicks, 0.4810 when counts can only use clicks from before the test period (only 3.3% of candidates keep a non-zero 24 h count) and 0.4854 with none — below chance. We therefore submitted the best models that need no click logs: NRMS on MIND and NRMS + freshness (publish times are in the test files) on EB-NeRD. The inference path was verified before submitting: on MIND_{small_dev} it reproduces the offline AUC (0.6232; Microsoft’s official `evaluate.py` gives 0.6235 on the written ranks, the difference being rank ties), and on EB-NeRD small validation 0.5539 / 0.6459; unseen title tokens use pretrained XLM-R vectors (2.9% of MIND and 15.8% of EB-NeRD test titles), and every zip is validated line by line against the source file (row order, ids, rank permutations). Rank-averaging click-free scorers looks promising offline (MIND NRMS + A1 semantic 0.6531; EB-NeRD NRMS + freshness with exposure counts 0.6609) but was not submitted.

Table 11: Codabench submissions (hidden test sets).

competition	model	impressions	zip MB	offline AUC	AUC	MRR	nDCG@5	nDCG@10
MIND 13967	NRMS	2,370,727	107.3	0.6232			<i>pending</i>	
EB-NeRD 2469	NRMS + freshness	13,536,710	221.4	0.6459			<i>pending</i>	

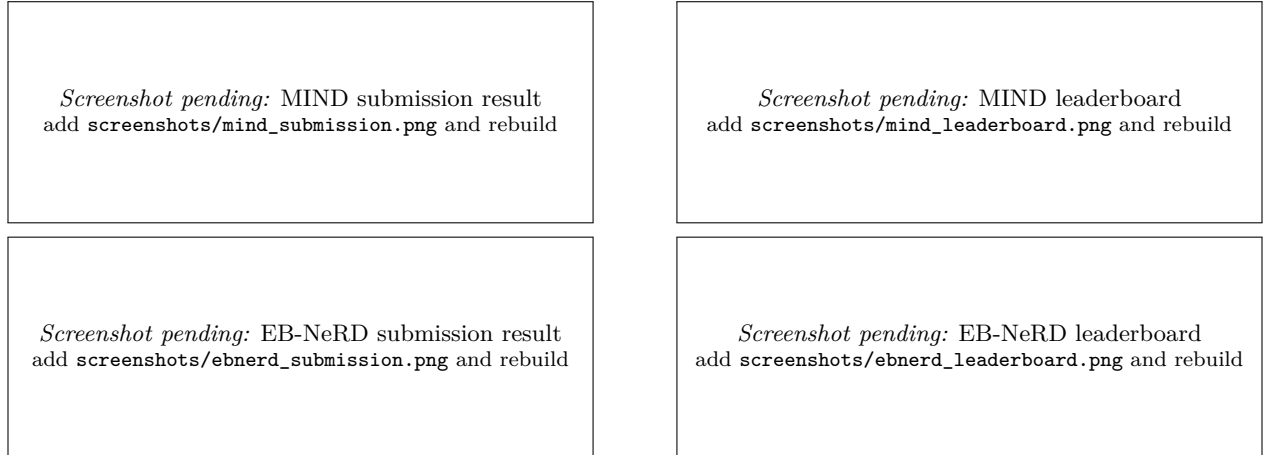


Figure 1: Codabench: submission results and leaderboards (MIND top, EB-NeRD bottom).

6 Q4: Serving and scale analysis

A one-request serving path loads everything once (indexes, model, every article’s NRMS vector, so a request never encodes text) and times eight stages; it reproduces the offline test scores (max difference 1e-06 MIND, 4e-06 EB-NeRD). PyTorch, FAISS and BLAS are pinned to one thread (they defaulted to 4–10), so figures are per core.

Table 12: Q4 serving: full two-stage pipeline, one request at a time, Apple M4 CPU, one thread; 2,000 replayed test requests.

	MIND	EB-NeRD small
memory to serve (MB)	446	169
largest: A1 / NRMS article vectors, NRMS weights (MB)	100 / 104 / 68	25 / 33 / 44
ANN index, catalogue: exact FAISS → HNSW (MB)	100 → 118	25 → 31
latency p50 / p95 / p99, one request (ms)	1.14 / 1.77 / 2.24	1.74 / 3.34 / 3.81
same, calling batch code per request (ms)	26.1 / 28.5 / 29.6	9.6 / 11.1 / 11.7
cores at 1k / 10k QPS (sim. p99 ms)	3 / 21 (2.2)	4 / 32 (3.8)
cost per 1,000 queries at 1k QPS	\$0.000033	\$0.000044

Findings. BM25 is the largest stage and sets the tail: its query is the user’s whole click history (EB-NeRD users with 200+ clicks: p99 3.96 ms vs 1.06 ms for 1–10). NRMS re-ranking is a constant ~ 0.3 ms: 200 dot products over precomputed vectors. Calling A1’s batch functions per request is $23.0\times$ slower on MIND, almost all of it rebuilding an article→tokens dictionary over the whole catalogue per call; caching request-independent work removes it. **Cost** (assumed \$0.04/core-hour, one single-threaded worker per core, 60% utilisation): a queueing simulation driven by the measured request times (validated against M/M/1 theory) keeps p99 under 10 ms even at 95% utilisation, so the 100 ms SLA never binds and cost is linear in traffic; a $3\times$ slower cloud vCPU still meets it. Network, parsing and remote stores are not in these timings.

7 Where the system breaks at $10\times$

Measured on a synthetic $10\times$ catalogue (every article copied $10\times$ with perturbed vectors; MIND 65,238 → 652,380 articles) and $10\times$ click events, replaying 1,000 real users:

- **First: per-request code that touches the whole catalogue.** Calling the batch code per request, MIND p99 goes from 28 to **308 ms** and breaks the SLA on a single request (EB-NeRD 91 ms). With request-independent work cached, serving p99 grows only $1.8 \rightarrow 4.9$ ms.
- **Second: RAM per worker.** Served memory grows $446 \rightarrow 3,844$ MB; each MIND worker needs 4.2 GB, above an assumed 4 GB per vCPU, so RAM, not CPU, sets the bill (60 vCPUs for 57 cores of work at $10\times$ catalogue and traffic). Fix: share read-only vectors across workers (memory-mapping) or move them to a vector-search service.
- **Third: approximate-search quality.** Exact search over the whole MIND catalogue reaches 18 ms p99 at $10\times$, pushing a larger site to HNSW [8, 9], whose recall@100 falls from 0.81 to 0.46 (efSearch 64; 0.73 at 256). Searching only the circulating pool avoids this today (1.4 ms) but the pool grows with the site.
- **Not the bottleneck:** stage-2 re-ranking and click-count lookups stay flat; $10\times$ traffic alone is linear cost (lower p99 from larger worker pools). Argued, not measured: $10\times$ click events need a streaming counter, and the popularity model’s accuracy depends on that counter being real-time (it collapses to chance without it, §5).

Caveats: laptop CPU (indicative per-core numbers); synthetic near-duplicate copies make HNSW recall pessimistic and do not grow BM25’s vocabulary; users $\times 10$ is a linear memory extrapolation.

8 Anti-gaming, limitations and alternatives

Features unavailable at serving time (Q9). EB-NeRD ships lifetime totals (total_inviews, total_pageviews, total_read_time), known only after the fact. Adding them to the A2 GBDT’s

features inflates every metric significantly (Table 13); using `total_pageviews` alone as a popularity score inflates A1’s popularity scorer far more, 0.5978 against 0.4459 AUC (+0.1520). The A2 feature table excludes all of them by default (asserted by `tests/test_no_leakage.py`), and every trailing count is strictly causal. The Codabench analysis (§5) is the serving-time version of the same lesson: a feature that is honest offline can still be unavailable at serving time if the live system does not log it.

Table 13: Q9 on the A2 re-ranker (EB-NeRD small, 10,000 test impressions): the same GBDT trained with the dataset’s serving-time-unavailable article totals added. * = CI excludes zero.

metric	honest (A2 features)	with leaked features	inflation, 95% CI
auc	0.7499	0.7605	+0.0105 [+0.0074, +0.0136]*
mrr	0.5226	0.5315	+0.0089 [+0.0038, +0.0141]*
ndcg@5	0.5831	0.5954	+0.0123 [+0.0077, +0.0169]*
ndcg@10	0.6208	0.6288	+0.0080 [+0.0040, +0.0121]*

Limitations. MIND’s gain is within seed noise; Q3 runs mix Apple-GPU and CUDA hardware across seeds; the circulating candidate pool is built from the whole test period (slightly optimistic recall); Universe B labels a retrieved but never-shown article as a non-click; Q5 uses 3,500 impressions and reports coverage without a CI.

Alternatives considered. A cross-encoder or full transformer news encoder (too slow to train on a laptop in the time available; NRMS epochs took 11–13 min on an M4); retraining the GBDT on retrieved rather than shown negatives (would fix the MIND distribution shift in Universe B); exposure counts as a click-free popularity proxy for Codabench (0.636 AUC alone on EB-NeRD, measured but not submitted); HNSW with larger efSearch for a bigger catalogue.

References

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- [9] J. Johnson, M. Douze, H. Jégou. Billion-scale similarity search with GPUs. IEEE Trans. Big Data 2021.

A Reproduce

```
make data # A1 pipeline; EB-NeRD at scale: small
python src/rerank/evaluate_reranker.py --config config/<ds>.yaml \
    --sample 10000 --retrieve-sample 3000 # Q2
make q3 # Q3: 18 NRMS runs (resumable) + tables
make q4 # Q4: serving, memory, latency, cost, 10x
python src/eval/evaluate_twostage.py --config config/<ds>.yaml --sample 3500 # Q5
python src/submission/predict_nrms.py ... # Codabench zips; validate_zip.py checks them
python src/report/build_design_note.py # this document
```

Detailed tables are in `reports/q3_summary.md` and `reports/q4_summary.md`; each question’s implementation notes are in `reports/a2_q*_implementation.md`. AI assistance is logged per question in `reports/a2_ai_usage_log.md` (A1’s is `reports/ai_usage_log.md`); the run logs behind every number are in `logs/`. The build also writes three figures to `reports/design_note/figures/`, omitted here for length.